SUKKUR IBA UNIVERSITY

MERIT – QUALITY – EXCELLENCE

<An Artificial Intelligence Based Agronomist>

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Outline

- Introduction of Project
- Motivation
- Problem Statement
- Literature Review
- Aims and Objectives
- Methodology
- App Development
- Achievements

PROJECT INTRODUCTION

- We have developed a Mobile App
- Plant Identification using Images
- Plant Information
- Using AI

MOTIVATION

- Providing Awareness to People
- Green Pakistan
- Promoting Home Gardening
- Helping Farmers

PROBLEM STATEMENT

- Its very difficult to identify plants without help of any relevant person.
- It needs a smart solution.
- Though some same applications are in market but they provide region limited solutions.

LITERATURE REVIEW

- Promoting Home gardening Helps human health.

[Mohsin, M., Anwar, M. M., Jamal, F., Ajmal, F., & Breuste, J. (2017). Assessing the role and effectiveness of kitchen gardening toward food security in Punjab, Pakistan: a case of district Bahawalpur. *International Journal of Urban Sustainable Development*, *9*(1), 64-78.]

- Kitchen Gardening Helps people to Produce Healthy Food.

[Bajwa, B. E., Aslam, M. N., & Malik, A. H. (2015). Food security and socio-economic conditions of women involved in kitchen gardening in Muzaffargarh, Punjab, Pakistan. *Journal of Enviornmental and Agricultural Studies*, *4*.]

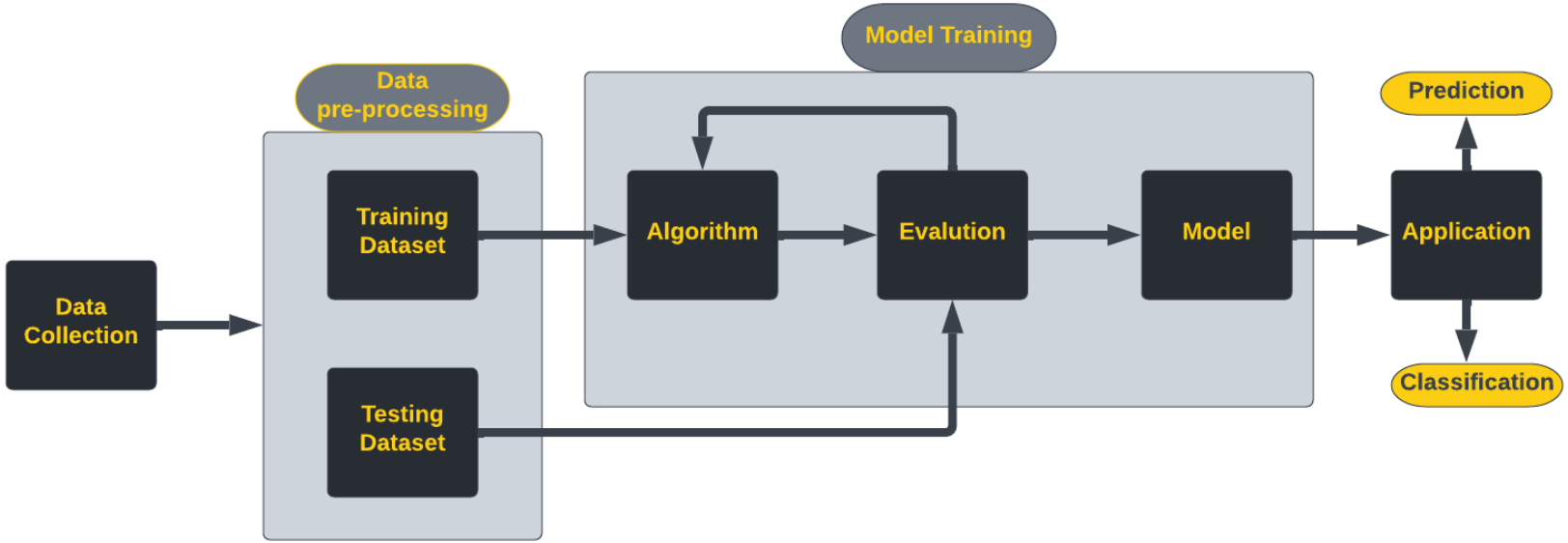
- This study proposes the use of CNNs and transfer learning to improve flower cognitive ability.

[Murugeswari, R., Nila, K. V., Dhananjeyan, V. R., Teja, K. B. S., & Prabhas, K. V. (2022, January). Flower perception using Convolution Neural Networks based Escalation of Transfer learning. In *2022 4th International Conference on Smart Systems and Inventive Technology (ICSSIT)* (pp. 1108-1113). IEEE.]

Achievements

- DICE AFS 2021 MNS Multan (Winner, PKR#-\ 1,00,000)
- Hult Prize 2022 SIBAU (2nd Runner-up)
- Lab2Market PAF IAST Hari Pur (Top 5th)
- Ideagist Pakistan 2021 (City Round Qualifier)

Flow Chart


Model Training
Data pre-processing
Prediction
Training Dataset
Algorithm
Evalution
Model
Application
Data Collection
Testing Dataset
Classification

METHODOLOGY

Data Collection

- Capture the images using mobile phones
- At different time and angle
- 20 classes
- Each class (Approximately 230 images)
- (Approximately 4600 images)

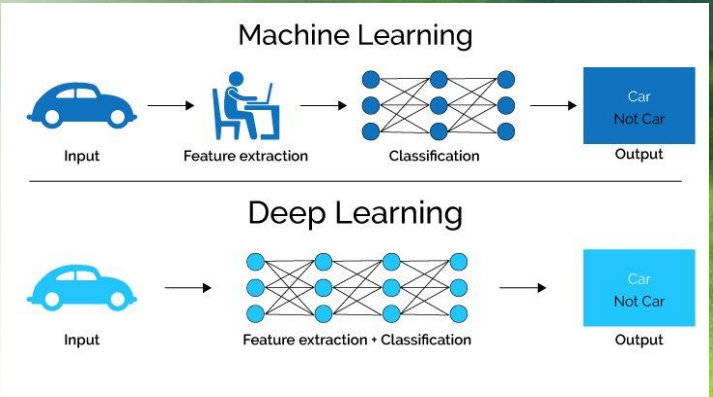
Pre processing

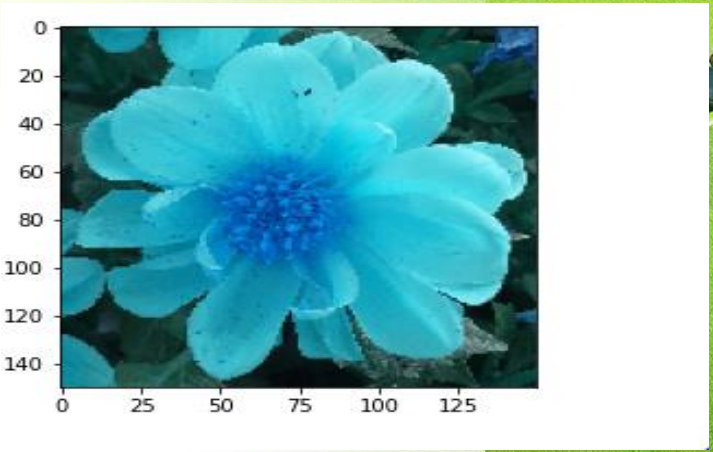
- Done manually cropping

METHODOLOGY Conti…

Model Training

- ❑ CNN
- ❑ Dataset
  - Classes (20)
    1. Train (160)
    2. Test(10)
    3. Validation (30)
- ❑ Load dataset
- ❑ Resize (150,150)
- ❑ Labeling
- ❑ Normalizing by 255





Build AI Model

Uncropped Data (10 Classes)

- model.Sequential()

Feature Extraction

- Pre-trained Model (VGG-16)
- 13 conv layers, 13 Max pooling and 3 dense layers

Classification

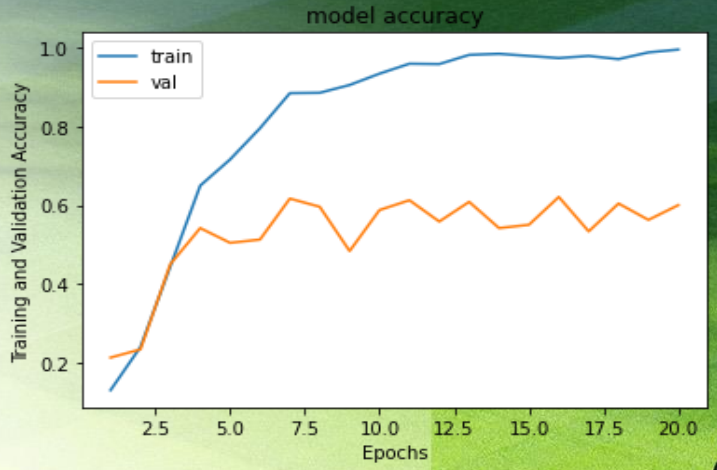
- Final output layer with 10 neurons (softmax activation)

Compiling the model

- Model.compile
- Adam optimizer (0.0001)

Training the model

- Model.fit (20 epochs)
- Accuracy 47% Training



Cropped Data (10 Classes)

- model.Sequential()

Feature Extraction

- 3 convolutional 2D layers with 128,512 and 1028 filters of 3x3 size (Relu activation)
- Max_pooling 2D (3X3)
- Dropout layer (0.4)

Classification

- Flatten layer
- 1 Dense layer is used with 512 neurons (Relu activation)
- Final output layer with 10 neurons (softmax activation)

Compiling the model

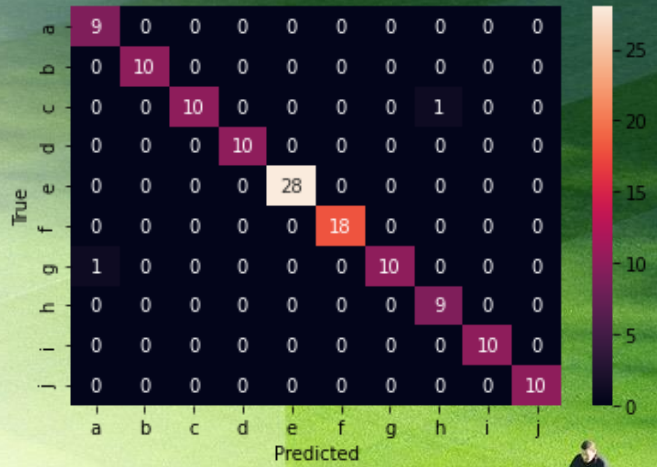
Model.compile

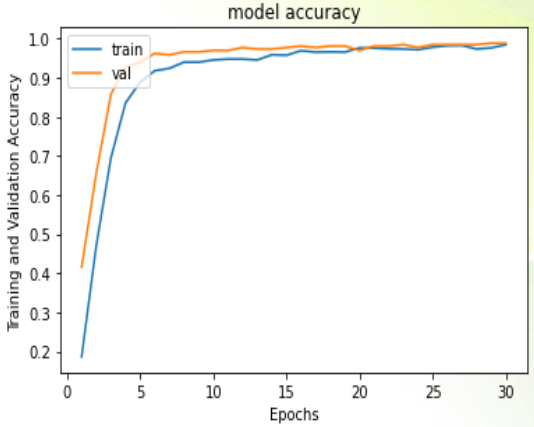
Adam optimizer (0.00001)

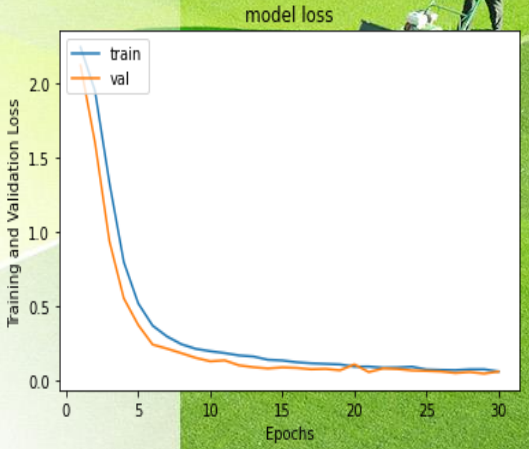
Training the model

Model.fit (30 epochs)

Accuracy 97.6% Training, 98% Validation







Cropped Data (20 Classes)

- model.Sequential()

Feature Extraction

- 2 convolutional 2D layers with 32 and 64 filters of 3x3 size (Relu activation)
- Maxpooling 2D (3X3)
- Dropout layer (0.6)

Classification

- Flatten layer
- 1 Dense layer is used with 512 neurons (Relu activation)
- Final output layer with 20 neurons (softmax activation)

Compiling the model

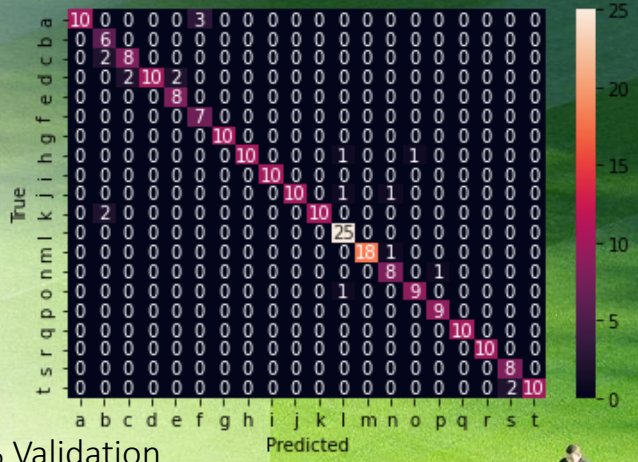
Model.compile

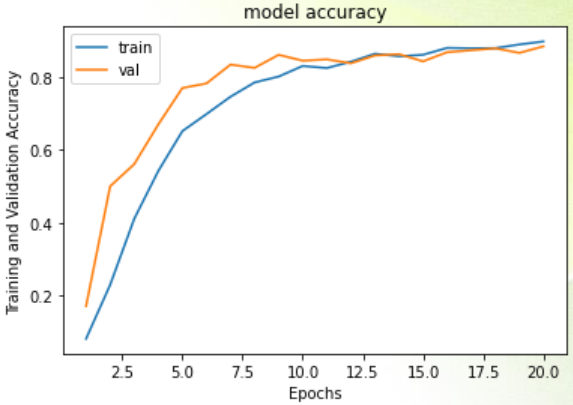
Adam optimizer (0.00001)

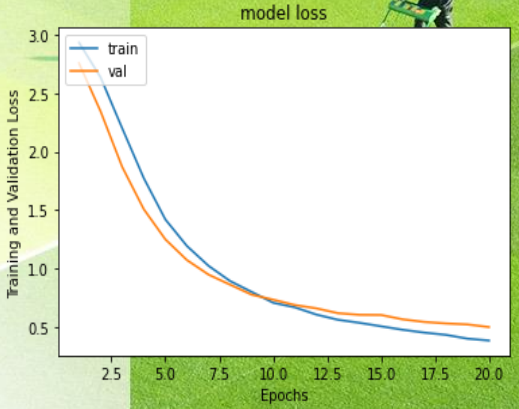
Training the model

Model.fit (20 epochs)

Accuracy 92% Training, 91% Validation







APP DEVELOPMENT

Features

- Plant Identification

Future Work:

- Plant Disease Detection
- Crop Identification
- Shopping
- Consultation

Requirements

- External Libraries
- Introduction_Screen
- Image_picker
- Translator
- Image_cropper
- hexcolor

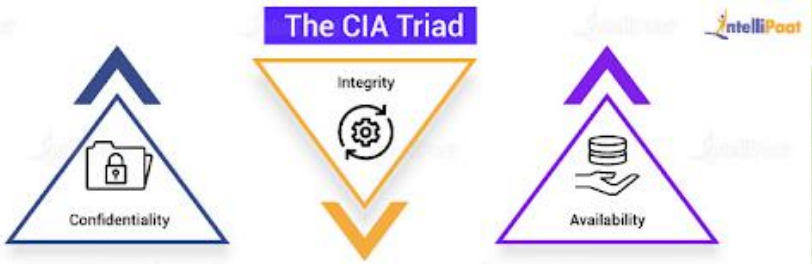
Platform

- Frontend is developed on Flutter with Dart.
- We are using Python Flask webserver as Backend
- Using Azure Microsoft Server hosting services

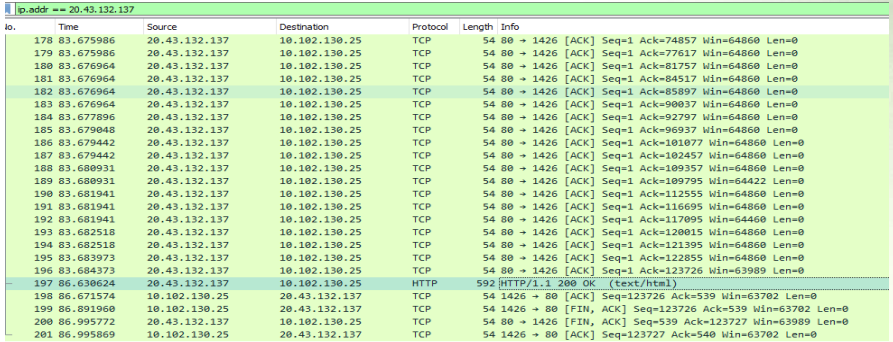
Security of App, Request & Response and Server

Implemented CIA Triad

- **Confidentiality** (HTTP Vs HTTPS)
- **Integrity** (Implemented by Azure)
- **Availability** (maintained by buying a premium package of Azure)



Responses



No.	Time	Source	Destination	Protocol	Length	Info
178	83.675986	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=74857 Win=64860 Len=0
179	83.675986	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=77617 Win=64860 Len=0
180	83.676964	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=81757 Win=64860 Len=0
181	83.676964	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=84517 Win=64860 Len=0
182	83.676964	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=85897 Win=64860 Len=0
183	83.676964	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=90037 Win=64860 Len=0
184	83.677896	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=92797 Win=64860 Len=0
185	83.679048	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=96937 Win=64860 Len=0
186	83.679442	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=101077 Win=64860 Len=0
187	83.679442	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=102457 Win=64860 Len=0
188	83.680931	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=109357 Win=64860 Len=0
189	83.680931	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=109795 Win=64422 Len=0
190	83.681941	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=112555 Win=64860 Len=0
191	83.681941	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=116695 Win=64860 Len=0
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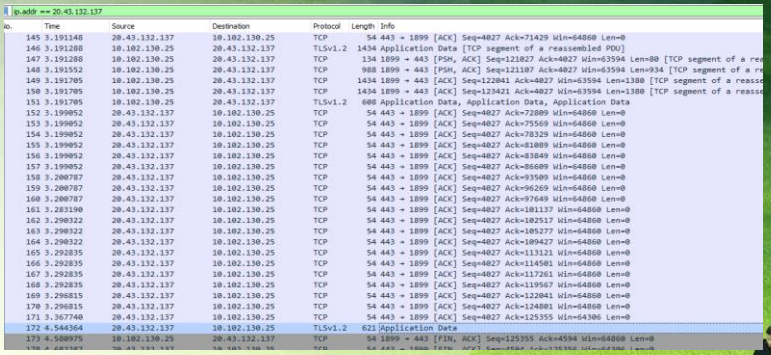
193	83.682518	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=120015 Win=64860 Len=0
194	83.682518	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=121395 Win=64860 Len=0
195	83.683973	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=122855 Win=64860 Len=0
196	83.684373	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [ACK] Seq=1 Ack=123726 Win=63989 Len=0
197	86.630624	20.43.132.137	10.102.130.25	HTTP	592	HTTP/1.1 200 OK (text/html)
198	86.671574	10.102.130.25	20.43.132.137	TCP	54	1426 → 80 [ACK] Seq=123726 Ack=539 Win=63702 Len=0
199	86.891960	10.102.130.25	20.43.132.137	TCP	54	1426 → 80 [FIN, ACK] Seq=123726 Ack=539 Win=63702 Len=0
200	86.995772	20.43.132.137	10.102.130.25	TCP	54	80 → 1426 [FIN, ACK] Seq=539 Ack=123727 Win=63989 Len=0
201	86.995869	10.102.130.25	20.43.132.137	TCP	54	1426 → 80 [ACK] Seq=123727 Ack=540 Win=63702 Len=0

Request Response Cycle Over HTTP



No.	Time	Source	Destination	Protocol	Length	Info
145	3.191148	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=71429 Win=64860 Len=0
146	3.191288	10.102.130.25	20.43.132.137	TLSv1.2	1434	Application Data [TCP segment of a reassembled PDU]
147	3.191288	10.102.130.25	20.43.132.137	TCP	134	1899 → 443 [PSH, ACK] Seq=121027 Ack=4027 Win=63594 Len=80 [TCP segment of a re
148	3.191552	10.102.130.25	20.43.132.137	TCP	988	1899 → 443 [PSH, ACK] Seq=121187 Ack=4027 Win=63594 Len=934 [TCP segment of a re
149	3.191705	10.102.130.25	20.43.132.137	TCP	1434	1899 → 443 [ACK] Seq=122041 Ack=4027 Win=63594 Len=1380 [TCP segment of a reasse
150	3.191705	10.102.130.25	20.43.132.137	TCP	1434	1899 → 443 [ACK] Seq=123421 Ack=4027 Win=63594 Len=1380 [TCP segment of a reasse
151	3.191705	10.102.130.25	20.43.132.137	TLSv1.2	608	Application Data, Application Data, Application Data
152	3.199052	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=72809 Win=64860 Len=0
153	3.199052	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=75569 Win=64860 Len=0
154	3.199052	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=78329 Win=64860 Len=0
155	3.199052	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=81089 Win=64860 Len=0
156	3.199052	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=83849 Win=64860 Len=0
157	3.199052	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=86609 Win=64860 Len=0
158	3.200787	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=93509 Win=64860 Len=0
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161	3.283190	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=101137 Win=64860 Len=0
162	3.290322	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=102517 Win=64860 Len=0
163	3.290322	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=105277 Win=64860 Len=0
164	3.290322	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=109427 Win=64860 Len=0
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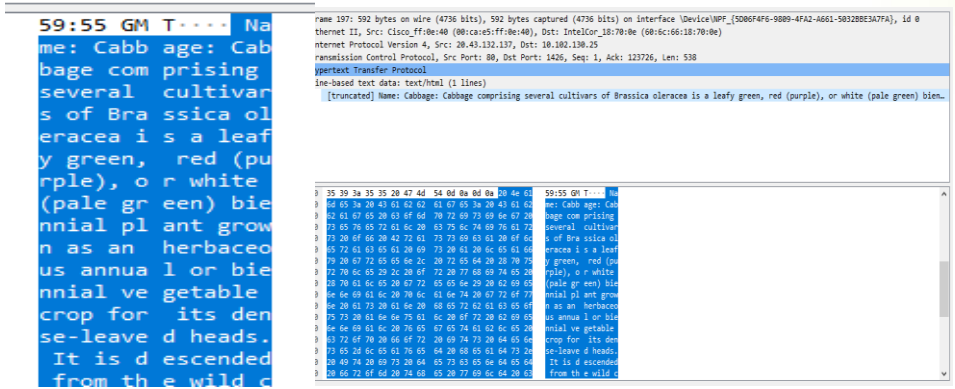
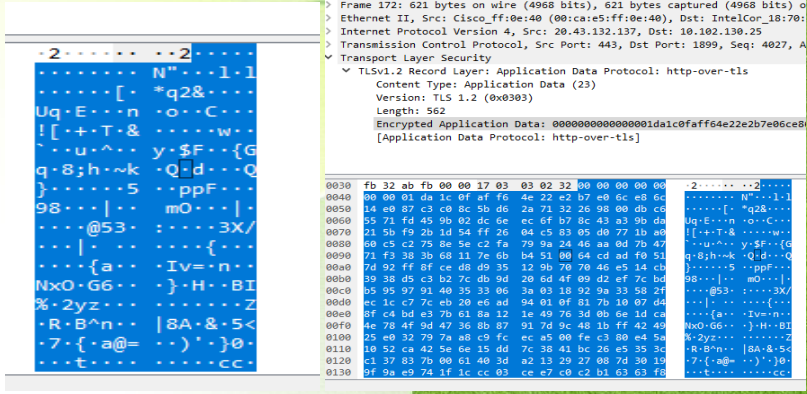
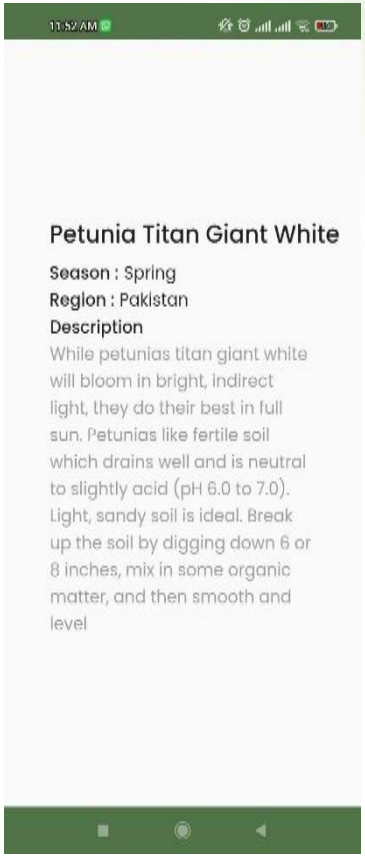
168	3.292835	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=119567 Win=64860 Len=0
169	3.296815	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=122041 Win=64860 Len=0
170	3.296815	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=124801 Win=64860 Len=0
171	3.367740	20.43.132.137	10.102.130.25	TCP	54	443 → 1899 [ACK] Seq=4027 Ack=125355 Win=64306 Len=0
172	4.544364	20.43.132.137	10.102.130.25	TLSv1.2	621	Application Data
173	4.580975	10.102.130.25	20.43.132.137	TCP	54	1899 → 443 [FIN, ACK] Seq=125355 Ack=4594 Win=64860 Len=0

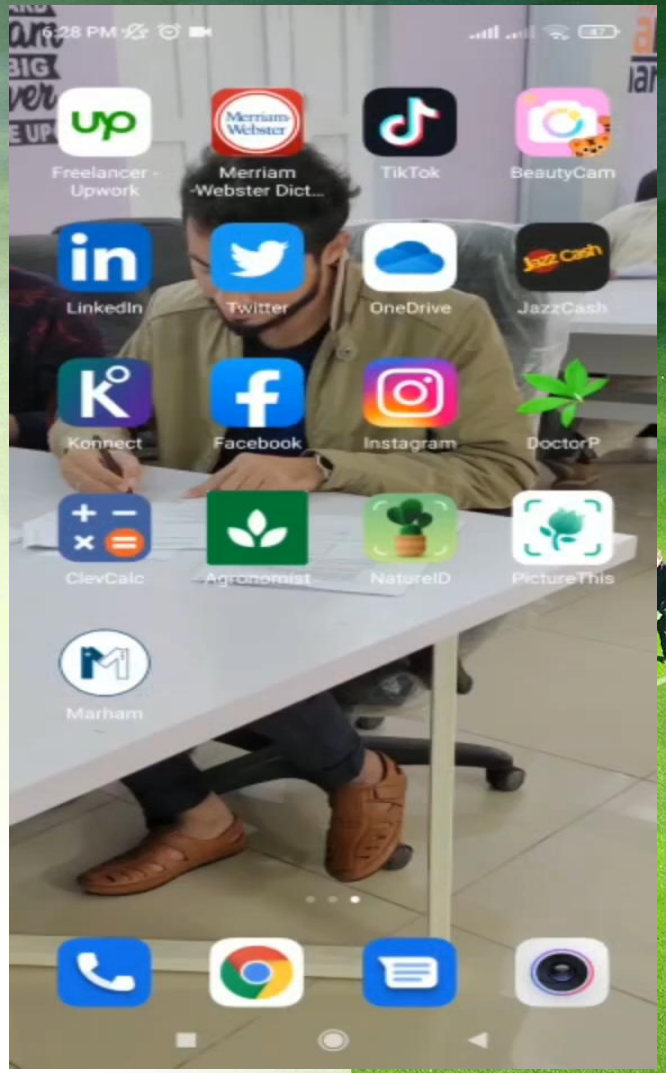
Request Response Cycle Over HTTPS





How It Works and Results





CONCLUSION

- Project Goal was to do identification of plants
- Collected the data set of 20 classes of flower Plants
- Applied some pre-processing techniques
- Applied the convolutional neural networks as deep learning model to train the model
- Developed Mobile App
- Python based Flask Backend on Azure server
- Testing

FUTURE RECOMMENDATIONS

- Diseases and Crop Yield Prediction
- We have applied for ISF, a world bank grant through HEC.
- We will go for a national app.
- Start-up
- Employement

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Thank You!